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Figure 10.2. Motions

10.1.1 Periodic and non-periodic motion

Motion in physics can be classified as repetitive (periodic motion) and non-repetitive (non-periodic motion).

1. **Periodic motion**
   Any motion which repeats itself in a fixed time interval is known as periodic motion.
   Examples : Hands in pendulum clock, swing of a cradle, the revolution of the Earth around the Sun, waxing and waning of Moon, etc.
2. **Non-Periodic motion**
   Any motion which does not repeat itself after a regular interval of time is known as non-periodic motion.
   Example : Occurance of Earth quake, eruption of volcano, etc.

EXAMPLE 10.1

Classify the following motions as periodic and non-periodic motions?.

a. Motion of Halley's comet.
b. Motion of clouds.
c. Moon revolving around the Earth.

Solution

a. Periodic motion
b. Non-periodic motion
c. Periodic motion

EXAMPLE 10.2

Which of the following functions of time represent periodic and non-periodic motion?.

a. $\sin \omega t + \cos \omega t$
b. $\ln \omega t$

Solution

a. Periodic
b. Non-periodic

Question to ponder

Discuss "what will happen if the motion of the Earth around the Sun is not a periodic motion".

10.1.2 Oscillatory motion

When an object or a particle moves back and forth repeatedly for some duration of time its motion is said to be oscillatory (or vibratory). Examples; our heart beat, swinging motion of the wings of an insect, grandfather's clock (pendulum clock), etc. Note that all oscillatory motion are periodic whereas all periodic motions need not be oscillation in nature. see Figure 10.3